1. Clinical Question & Core Hypothesis

Clinical Question

Obesity elevates leptin (LEP), yet it is unclear why some TP53-mutant/TNBC tumors in obese patients are particularly chemoresistant and aggressive despite comparable tumor cell–intrinsic features. 2) Tier 1 paths used: (a) Obesity → ↑LEP → impacts breast cancer biology; (b) TP53 mutated in breast/TNBC and linked to aggressive phenotype and therapy resistance. I built on these because they form a direct host–tumor interaction axis (LEP ↔ TP53 status) that can mechanistically modify treatment response. 3) Tier 2 evidence supporting the claim: adipokine-driven stromal remodeling and ECM (NABA_CORE_MATRISOME genes such as COL1A1, FN1) and proliferation (MKI67) can increase stromal-to-tumor ratio and impede drug delivery; leptin receptor signaling (LEPR) provides the signaling conduit. These support a model where LEP-induced desmoplasia cooperates with TP53 loss to blunt chemotherapy. 4) Choice of endpoint: I selected therapy_response because the input emphasizes chemotherapy regimens and the clinical problem is chemo-resistance in obese/TP53-mutant TNBC — therapy response is the most directly relevant and actionable endpoint. 5) Deliberate exclusions: I excluded broader immune-centric imaging features (TILs/TLS) and many other pathway genes to keep the hypothesis sharply focused on stromal remodeling as the mediating mechanism; other immune- or metabolism-related genes are placed in secondary_modifiers for later exploration?

Working Hypothesis

In TP53-mutant triple-negative breast cancer, obesity-associated elevation of leptin (LEP) drives stromal remodeling via LEPR signaling in the tumor microenvironment, increasing the stromal-to-tumor ratio (TSR) through upregulation of ECM components COL1A1 and FN1 and related stromal activity. This high-TSR, desmoplastic milieu impedes chemotherapy penetration and promotes tumor cell survival, resulting in poorer therapy_response. The effect is amplified in tumors with high LEP/LEPR signaling and TP53 loss-of-function, and can be mitigated by disrupting LEPR signaling or modulating the ECM. This leads to testable predictions: (1) TP53-mutant TNBC with high LEP/LEPR expression and high TSR will exhibit reduced chemotherapy response; (2) blocking LEPR signaling or ECM deposition will restore chemosensitivity; and (3) imaging of TSR can serve as a predictive biomarker of therapy_response in this context. The investigation will focus on two anchor genes (TP53, LEP), with secondary modifiers (LEPR, COL1A1, FN1, MKI67) to reflect stromal remodeling, proliferation, and signaling, while using TSR as the primary imaging-endpoint and therapy_response as the clinical readout.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	TP53, LEP
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TP53: TP53-mutant/TNBC tumors in obese patients are particularly chemoresistant and aggressive despite comparable tumor cell–intrinsic features.

LEP: leptin (LEP), yet it is unclear why some TP53-mutant/TNBC tumors in obese patients are particularly chemoresistant and aggressive despite comparable tumor cell–intrinsic features.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	LEPR	Supporting modifier from knowledge graph
Secondary	COL1A1	Supporting modifier from knowledge graph
Secondary	FN1	Supporting modifier from knowledge graph
Secondary	MKI67	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: capture signaling, ECM remodeling, and proliferation within the stroma. This yields a testable, mechanistic hypothesis that can be explored through in vitro co-culture models, in vivo obesity and TP53-mutant TNBC models, and retrospective clinical correlations to establish TSR as a predictive biomarker for chemotherapy response in this subset.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- LEPR: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- COL1A1: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.